

AI SPORTS INJURY RISK DETECTION



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The Problem

1 HUMAN ERROR

Manual posture analysis is highly prone to human error, leading to inconsistent assessments and unreliable feedback for athletes and coaches.

2 INEFFICIENCY

Traditional movement analysis is time-consuming and labor-intensive, causing delayed feedback that slows athlete performance improvement.

3 INJURY RISK

Late intervention due to delayed analysis increases injury risk.
Manual Analysis → Human Error → Delay → Injury Risk.

Proposed Solution



AI VIDEO ANALYSIS

Automated video analysis using AI and Google MediaPipe BlazePose for accurate pose detection.



REAL-TIME RISK PREDICTION

Real-time prediction of sports injury risk based on joint angles and movement patterns.



EXERCISE RECOMMENDATION

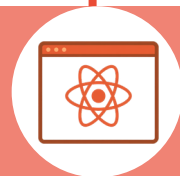
Personalized exercise recommendations generated automatically based on AI movement analysis.



REPORTS & HISTORY TRACKING

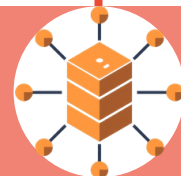
Comprehensive analysis reports with full history tracking stored in MongoDB dashboard.

Technology Stack



FRONTEND

React.js for dynamic UI components and Tailwind CSS for responsive, modern styling.



BACKEND

Node.js runtime with Express.js framework for RESTful API and server-side logic.



DATABASE & AUTH

MongoDB for data storage with JWT Authentication for secure user sessions.



AI PROCESSING

FFmpeg for video processing and Google MediaPipe BlazePose for pose detection.

System Architecture



Athlete
↓
React Frontend
↓
Node.js Backend

FFmpeg
↓
MediaPipe BlazePose
↓
Joint Angle Calculation
↓
Movement Analysis

Risk Prediction
↓
Exercise Recommendation
↓
Analysis Report
↓
MongoDB & Dashboard

Step-by-Step Workflow



LOGIN / SELECT ATHLETE



UPLOAD VIDEO / FRAME EXTRACTION



POSE DETECTION / JOINT ANGLE
CALCULATION



MOVEMENT ANALYSIS / RISK PREDICTION



EXERCISE RECOMMENDATION / REPORT /
SAVE HISTORY

Each step in the pipeline processes athlete video data progressively — from login to final report generation and history storage.

AI Working

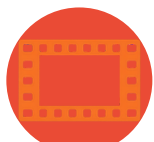


Video Upload → Frame
Extraction → 33 Body
Landmarks detected via
Google MediaPipe
BlazePose pose estimation
model.

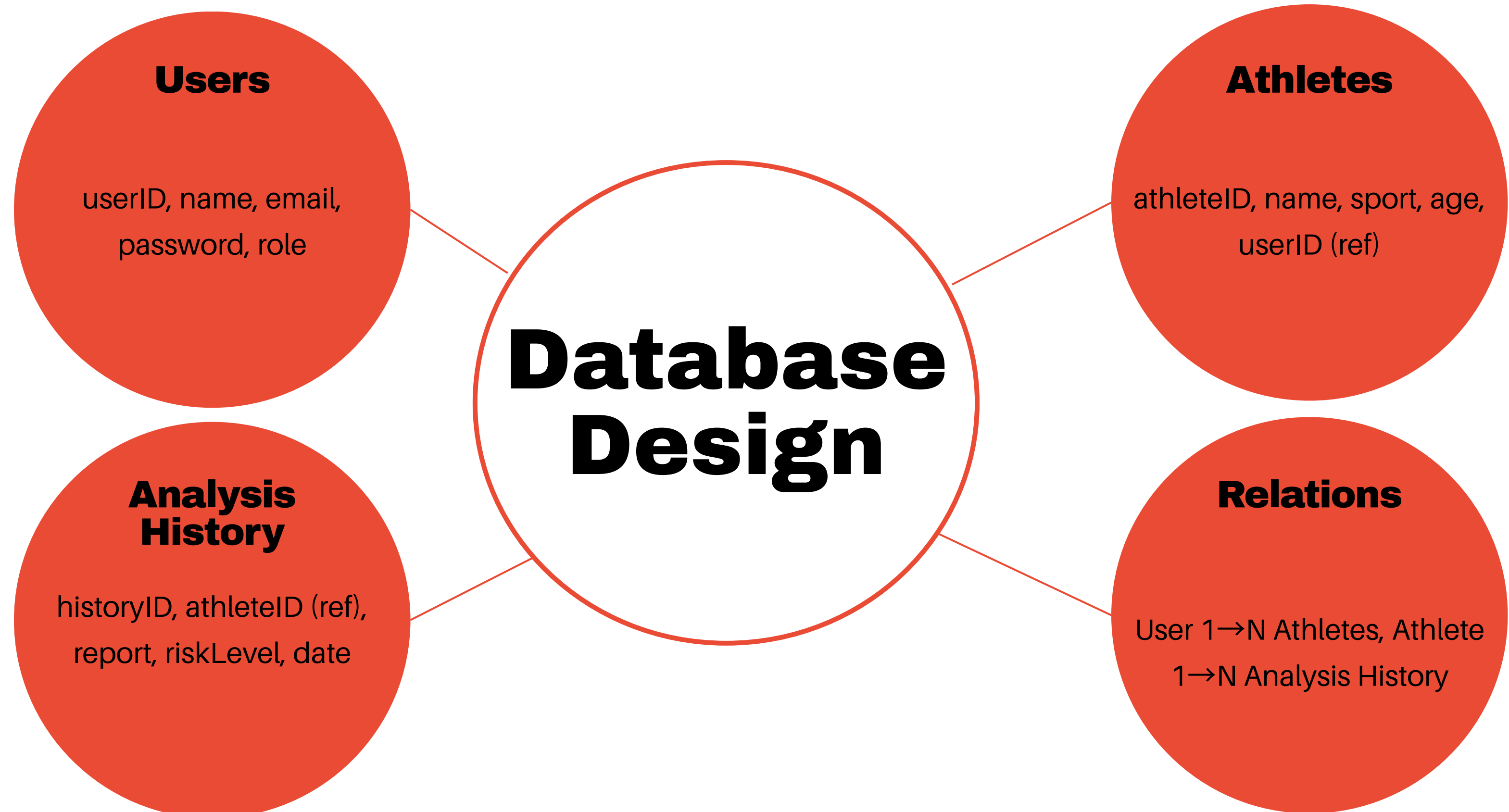
Joint Angle Calculation
using Vector Mathematics
→ Movement Analysis via
Rule-Based Algorithm →
Risk Prediction using
Decision Engine.

Personalized Exercise
Recommendation
generated →
Comprehensive Analysis
Report created → Results
saved to History &
Dashboard.

Algorithms Used

-  FFMPEG — VIDEO PROCESSING
-  MEDIAPIPE BLAZEPOSE — POSE DETECTION
-  VECTOR MATHEMATICS — JOINT ANGLE CALCULATION
-  RULE-BASED ENGINE — MOVEMENT ANALYSIS & RISK PREDICTION
-  TEMPLATE-BASED GENERATOR — ANALYSIS REPORT

Algorithm / Function
FFmpeg — Video Processing
MediaPipe BlazePose — Pose Detection
Vector Mathematics — Joint Angles
Rule-Based Engine — Risk Prediction
Template Generator — Reports



Project Features



AUTHENTICATION & ATHLETE MANAGEMENT

JWT-based secure login, user registration, and complete athlete profile management with role-based access.



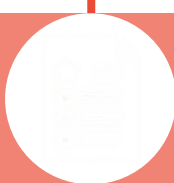
VIDEO UPLOAD & POSE DETECTION

Upload sports videos via FFmpeg; MediaPipe BlazePose detects 33 body landmarks for full pose analysis.



MOVEMENT ANALYSIS & RISK PREDICTION

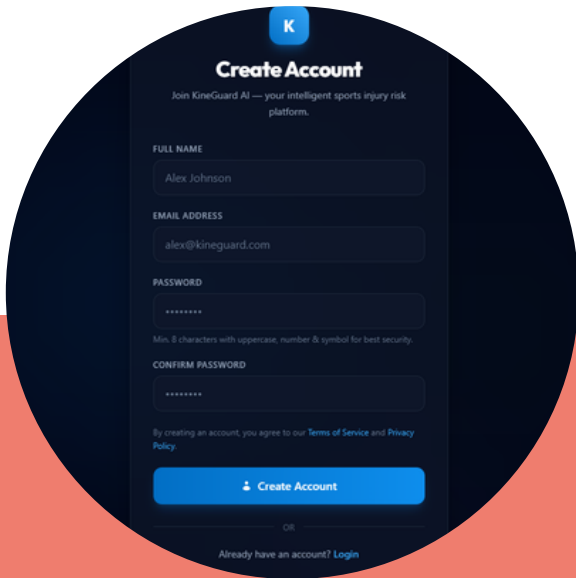
Joint angle calculation, rule-based movement analysis, and automated sports injury risk prediction engine.



RECOMMENDATIONS & REPORTS

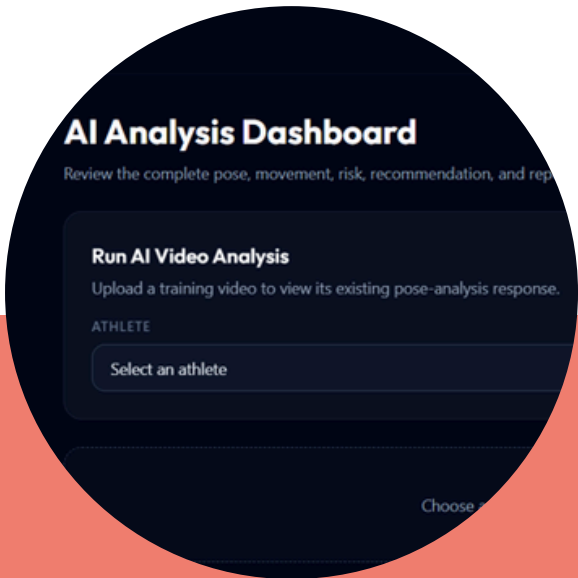
Personalized exercise recommendations, detailed analysis reports, history tracking, and athlete dashboard.

Project Results



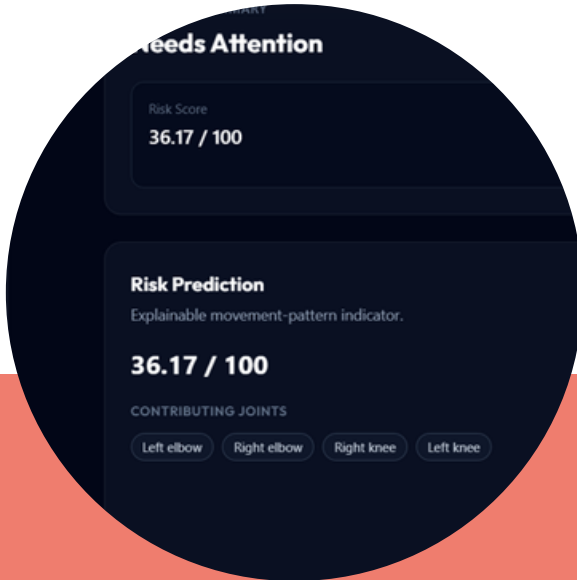
LOGIN & REGISTER

Registered users can log in with their email and password.



DASHBOARD

main analytics dashboard view



ANALYSIS HISTORY & FINAL REPORT

Historical records view and comprehensive final report output

Challenges & Future Scope

CHALLENGES

- Lighting conditions affect pose detection accuracy
 - Camera angle variability impacts joint analysis
 - Video quality issues reduce landmark precision
- Pose detection accuracy with complex movements
 - Threshold tuning difficulties for risk rules

FUTURE SCOPE

- Integrate machine learning models for smarter prediction
 - Live camera feed analysis in real-time
 - PDF report export feature for coaches
 - Athlete progress tracking over time
 - Dedicated coach dashboard



Conclusion



FFMPEG & MEDIAPIPE BLAZEPOSE FOR VIDEO PROCESSING AND POSE ANALYSIS



RULE-BASED AI ENGINE FOR MOVEMENT AND INJURY RISK ASSESSMENT



AUTOMATED PERSONALIZED EXERCISE RECOMMENDATION GENERATION



COMPREHENSIVE REPORT CREATION AND HISTORY MANAGEMENT



SCALABLE MERN STACK APP ENHANCING SPORTS INJURY PREVENTION



This project delivers an end-to-end AI-powered solution for real-time sports injury risk detection, combining modern web technologies with advanced pose analysis to empower athletes and coaches.

Thank You!

Sanskar Chaurasia



AI Sports Injury Risk Detection System



MERN Stack & Google MediaPipe BlazePose



This project was developed as part of Infosys Springboard Virtual Internship 7.0. We welcome any questions or feedback regarding the AI Sports Injury Risk Detection and Movement Analysis System.

